

Human Natural Structure

Analytical Specification

Inference-Level Causal Verification and the 864-Cell Analytical Engine

Volume 3 · Human Natural Structure Series

Satoru Hara

Natural Structure Works (NSW)

A Note on Method and Evidence

This volume specifies the analytical layer of Human Natural Structure, the 864 engine. Because it makes mathematical claims, it is important to separate three registers of assertion, which are kept distinct throughout.

Established results. The mathematics of the inference operators is Judea Pearl's: the causal hierarchy of association, intervention, and counterfactual; the do-operator and its identification by the back-door and front-door criteria; and the Causal Hierarchy Theorem, which shows the rungs do not collapse. The control operators rest on William Powers' Perceptual Control Theory and on classical negative-feedback control. The formal properties of the attenuation map are elementary real analysis. None of this is in dispute, and none of it is original to HNS.

HNS design decisions. What is original — and therefore a design choice rather than a theorem — is the assignment of exactly these six items as the third axis O1–O6; the multiplicative nesting $36 \times 4 \times 6 = 864$; the definition of the structural grounding score; the output-attenuation operator O6 and its use as a pre-decode actuator; and the use of the whole apparatus as an external verifier. These are engineering commitments, defensible but not proven by the borrowed mathematics that supports their parts.

Conjecture. Claims that the engine improves the reliability of a generative system in practice are hypotheses for empirical work, reported in Volume 1 and to be tested further.. They are flagged as conjecture wherever they appear and are not established by this specification.

Two honesty notes carried from the foundational volume apply again here. First, the recurrence of the number six across the framework is a chosen resolution, not a natural constant; the inference family has a principled three (Pearl's rungs) and the control family a principled three (reference, comparator, actuator), and their sum is six by composition, not by necessity. Second, the scholars named as lineage did not design HNS; their work is acknowledged as inspiration, and the recombination is the author's responsibility.

Contents

A Note on Method and Evidence	2
Contents	3
Preface: From Coordinate to Engine	6
Part I	7
1. The 864 Analytical Configuration	8
1.1 Definition and cardinality	8
1.2 The engine acts on positioned claims	8
1.3 Notation and conventions	9
1.4 A configuration, read aloud	9
2. The Inherited Coordinate	10
2.1 The six Natural Layers (L1–L6)	10
2.2 The six Abstract Categories (C1–C6)	10
2.3 The four Onto-Modal Quadrants (Q1–Q4)	10
2.4 Why the factors multiply	11
3. The Third Axis: Inference & Control Operators	12
3.1 Why operators, and why these	12
3.2 Division of labour between the families	13
3.3 Categories and operators	13
Part II	14
4. O1 — Conditioning	15
4.1 Formal definition	15
4.2 What conditioning licenses	15
4.3 What conditioning does not license	15
4.4 Properties of the conditional distribution	15
4.5 Threats at the first rung	16
4.6 Measures of association	16
4.7 A positioned O1 claim (non-normative)	16
5. O2 — Intervention (the do-operator)	18
5.1 The do-operator	18
5.2 Why intervention differs from conditioning	18
5.3 Identification	18
5.4 Role in the engine	19
5.5 The three rules of do-calculus	19
5.6 The front-door criterion	19
5.7 A numerical instance: Simpson’s paradox	20
5.8 Randomization as the gold standard	20
5.9 A back-door adjustment (worked, non-normative)	20
5.10 A graph and an adjustment (worked, non-normative)	21
6. O3 — Counterfactual	22

6.1 Formal content.....	22
6.2 The hierarchy does not collapse	22
6.3 Role in the engine.....	22
6.4 Structural causal models	23
6.5 The three-step evaluation.....	23
6.6 Probabilities of causation	23
6.7 A probability of necessity (worked, non-normative).....	23
6.8 A counterfactual on a toy model (worked, non-normative).....	24
7. Unsupported Causality	25
7.1 Definition.....	25
7.2 Detection rule	25
7.3 Worked examples.....	25
7.4 The analytical failure types and their mapping.....	26
7.5 The operator order is strict.....	27
7.6 Vocabulary cues for the asserted operator	27
7.7 Sub-forms of Unsupported Causality	27
7.8 Modality and the licensed operator.....	28
Part III	30
8. 04 — Reference Signal.....	31
8.1 The reference invariant	31
8.2 Setting the reference.....	31
9. 05 — Comparator and Error Detection	32
9.1 The control error.....	32
9.2 Relation to predictive coding	32
9.3 The feedback law.....	32
9.4 Hierarchies of control.....	33
9.5 Convergence of the loop (worked, non-normative)	33
10. 06 — Output Attenuation.....	34
10.1 The grounding score	34
10.2 The attenuation map	34
10.3 Zero-Incidence Attenuation	34
10.4 Placement and the meeting of the families	35
10.5 Shapes of the attenuation map.....	35
10.6 Composing the grounding score.....	35
10.7 Acting before the decode.....	36
10.8 A worked attenuation	36
10.9 The attenuation map tabulated	36
Part IV.....	38
11. The Generative Principle and Operator Accuracy.....	39
11.1 The generative principle.....	39
11.2 Operator accuracy	39

11.3 Operator trajectories	39
11.4 The single-operator rule and its limits	40
12. Architecture and Runtime Placement	42
12.1 Position among the three axes	42
12.2 Runtime form	42
12.3 Where Unsupported Causality is caught.....	42
12.4 The log record	43
12.5 Relation to the three axes, restated	43
12.6 What the engine does not do	43
13. Intellectual Lineage and Proponents	44
13.1 The inference family.....	44
13.2 The control family	44
13.3 What is HNS's own	44
13.4 The proponents, in brief.....	45
14. Empirical Status and Limitations	47
14.1 What is established.....	47
14.2 What is proposed.....	47
14.3 What remains conjecture	47
14.4 Threats to the account.....	47
14.5 Two further exposures.....	48
Appendix A — Normative Reference Tables.....	49
A.1 Inference & Control Operators (the third axis).....	49
A.2 The structural nesting.....	49
Appendix B — Configuration Template and Worked Configurations	50
B.2 Further worked configurations.....	50
Appendix C — Worked Examples.....	52
Appendix D — Do-Calculus and a Worked End-to-End Pass	54
D.1 The three rules, restated.....	54
D.2 A full pass.....	54
D.3 A multi-step reduction (sketch)	54
Appendix E — Notation Summary	56
Appendix F — Casebook.....	57
Glossary	60
References	62

Preface: From Coordinate to Engine

The foundational volume gave Human Natural Structure a place to stand. HNS-36 positions a proposition by causal origin and explanatory purpose; HNS-144 refines that position with the modal stance the claim takes — whether it speaks of the subjective or the objective, the contingent or the necessary. Both are acts of location. Neither yet does anything.

This volume specifies what the framework does once a claim has been located. The 864 layer is the analytical engine: it applies, to a positioned and observed claim, one of six operators that together exhaust the ways a claim can be reasoned about and the ways an output can be controlled. Three of the operators are operations of inference — they ask what may be concluded, and on what evidential strength. Three are operations of control — they govern what is finally emitted. The coordinate says where a claim is; the engine says what is being done with it, and whether that doing is warranted.

The characteristic failure at this resolution has a precise name. Unsupported Causality is the assertion of a causal or counterfactual conclusion on the strength of evidence that supports only association. It is the single most common way a fluent system says more than it has earned. Pearl's ladder makes the error formal: a claim that climbs to the rung of intervention or counterfactual must carry support from that rung, not from the rung below. The 864 engine is, in large part, the machinery for detecting and refusing that overreach — and, through the attenuation operator, for ensuring that ungrounded content is scaled toward zero before it is ever emitted.

A reader who wants the formal core may read Part II (the inference operators) and Part III (the control operators) with Appendix A. Part I fixes definitions and recalls the inherited coordinate; Part IV places the engine in the larger architecture and states, candidly, what has and has not been shown.

Part I

The Analytical Engine — HNS-864

1. The 864 Analytical Configuration

HNS-864 is the highest-resolution layer of the structural axis. Where HNS-36 has thirty-six positions and HNS-144 has one hundred forty-four, HNS-864 has eight hundred sixty-four analytical configurations. Each configuration is the unit on which the engine operates.

1.1 Definition and cardinality

An analytical configuration is a four-tuple consisting of a Natural Layer, an Abstract Category, an Onto-Modal Quadrant, and an Inference & Control Operator:

$$(L_i \times C_j \times Q_k \cdot O_m), \text{ with } i, j, m \in \{1, \dots, 6\} \text{ and } k \in \{1, \dots, 4\}.$$

The first two coordinates are the HNS-36 cell; the first three are the HNS-144 cell; the operator is the third axis specified in this volume. The cardinality follows by multiplication of independent factors:

$$|L| \cdot |C| \cdot |Q| \cdot |O| = 6 \cdot 6 \cdot 4 \cdot 6 = 864.$$

Equivalently, $36 \times 4 = 144$, and $144 \times 6 = 864$. The 144 layer is the input to the engine; the operator axis multiplies it. This volume adds only the final factor of six; the first three are inherited from the foundational and observational specifications and are recalled in Chapter 2.

Note. The four quadrants come from two binary dimensions (subjective/objective $\times$ necessary/contingent), giving $2 \times 2 = 4$, not four separate factors. The operator axis is a single factor of six. Any reading that multiplies more than these four factors does not yield 864 and is not the specified system.

1.2 The engine acts on positioned claims

The order of application is fixed and is itself a normative requirement. A claim is first positioned in HNS-36, then observed in HNS-144, and only then analysed by the application of an operator. An operator applied to an unpositioned claim has no defined meaning, because the operator acts on the claim's coordinate, not on its surface words. This ordering is what distinguishes the engine from a free-floating inference procedure: the same sentence, positioned at two different cells, is a different object for the engine to act upon.

The distinction between the coordinate and the engine can be put simply. The coordinate is descriptive: it answers where and in what modal stance a claim sits. The engine is operational: it answers what inference is being drawn and what control is being exercised. A reliable system must keep the two aligned — the inference it draws must not exceed the evidence its position affords, and the output it emits must be attenuated in proportion to how well that alignment holds.

1.3 Notation and conventions

A configuration is written $(L_i \times C_j \times Q_k \cdot O_m)$. A reasoning chain is an operator trajectory: the ordered sequence of operators asserted across its steps, read together with the cell trajectory of the foundational volume. Probabilities are written $P(\cdot)$; the intervention distribution is written $P(\cdot \mid \text{do}(\cdot))$. The control error is written $e = r - p$, with r the reference and p the perception. Sections marked normative state requirements; sections marked informative or non-normative supply motivation and examples and impose no requirement. Unless stated otherwise, the definitions and rules of this volume are normative.

1.4 A configuration, read aloud

It helps to read a full configuration aloud once. Take “the interest-rate rise will curb inflation.” Its layer is macro-social and institutional (L6): the claim originates in monetary policy. Its category is transformation (C5): it is about an operation, a rate being changed to bring something about. Its quadrant is objective and contingent (Q3): it speaks of the world, and the relation could hold or fail. Its operator is intervention (O2): it asserts that setting the rate brings about a change in inflation. The configuration is therefore $(L6 \times C5 \times Q3 \cdot O2)$, and reading it this way already tells the verifier what to demand — an identification for the interventional claim — before a word of the content is judged true or false.

2. The Inherited Coordinate

This chapter recalls, compactly, the three inherited factors so that the present volume is self-contained. The normative source for all three is the foundational and observational specification; nothing here overrides it.

2.1 The six Natural Layers (L1–L6)

Layer	Name	Lineage
L1	Physical / Substrate	Physics; BFO material entity
L2	Cellular / Informational	Computational neuroscience
L3	Cognitive / System 1	Kahneman Type 1
L4	Operational / System 2	Kahneman Type 2; PFC control
L5	Micro-Social / Interaction	Habermas
L6	Macro-Social / Institutional	Institutional theory; BFO social object

The layers are causal strata: the layer at which a claim originates, not the layer at which its effects are seen. The engine treats the layer as the domain over which an operator ranges.

2.2 The six Abstract Categories (C1–C6)

Cat.	Name	Lineage
C1	Continuant / Entity	Aristotle; BFO continuant
C2	Occurrent / Process	Aristotle; BFO occurrent
C3	Relational / Boundary	Aristotle; Kant
C4	Quality / State	Kant
C5	Transformation / Operation	Pearl (do-operator)
C6	Meta-Level / Schema	Kant

Category C5, transformation, is of particular interest at this resolution: a claim positioned at C5 is already a claim about an operation, and so interacts directly with the inference operators of Part II.

2.3 The four Onto-Modal Quadrants (Q1–Q4)

Quad.	Dimensions	Reading
Q1	Subjective × Contingent	preferences, impressions
Q2	Subjective × Necessary	commitments, constitutive rules
Q3	Objective × Contingent	empirical fact, measurement
Q4	Objective × Necessary	definition, logical truth

The quadrant matters to the engine because the warrant a claim needs depends on its modal stance. An objective-contingent claim (Q3) is exactly the kind that invites a causal reading and therefore the kind most exposed to Unsupported Causality.

2.4 Why the factors multiply

The four factors compose multiplicatively because they are, by design, independent questions. Where a claim originates (layer) does not fix how it is read (category); how it is read does not fix the modality in which it is asserted (quadrant); and none of these fixes the operation it performs (operator). Each question can in principle be answered independently of the others, so the number of distinct answers is the product, not the sum, of the per-factor counts — $6 \times 6 \times 4 \times 6 = 864$. Were the factors correlated — if every institutional claim had to be a process asserted as an intervention — the later factors would carry no information and the space would collapse toward 36. The specification therefore treats independence as a design target: the coordinate is built so that any combination is expressible, and a combination that is rare in practice is information about the claim, not a defect of the grid.

3. The Third Axis: Inference & Control Operators

The operator axis has two families of three. The inference operators 01–03 are the three rungs of Pearl’s causal hierarchy; they govern what may be concluded from a claim and on what evidential strength. The control operators 04–06 are drawn from Perceptual Control Theory together with an HNS-defined output regulator; they govern what is finally emitted. The full axis is given in the table below and developed operator by operator in Parts II and III.

Op.	Name	Lineage	Formal content
01	Conditioning	Statistics / Pearl	$P(Y X)$ — association (Rung 1)
02	Intervention / do	Pearl	$P(Y \text{do}(X))$ — intervention (Rung 2)
03	Counterfactual	Pearl	$P(Y_x X', Y')$ — counterfactual (Rung 3)
04	Reference Signal	Powers (PCT)	reference invariant r
05	Comparator / Error	Powers (PCT)	control error $e = r - p$
06	Output Attenuation	HNS Engineering	output = $g(s) \cdot \text{raw}$, $g(0)=0$

3.1 Why operators, and why these

The coordinate locates a claim; it does not say what is being done with it. Yet the reliability of a generated statement turns less on where it sits than on what operation it performs: whether it merely reports an association, asserts the effect of an action, or imagines an alternative history; and, downstream, whether the system emits it, attenuates it, or suppresses it. These are exactly the operations the six operators name.

The choice of these six is principled within each family and compositional across them. Within inference, Pearl’s hierarchy is provably a strict three-level ladder: association, intervention, counterfactual, with each level expressing questions the level below cannot answer. Within control, the minimal negative-feedback loop

has exactly three functional parts: a reference that says what is wanted, a comparator that measures the gap, and an actuator that acts to close it. HNS adopts these two principled triples and composes them; the resulting six is a composition, not an independent postulate that the world contains six operators.

3.2 Division of labour between the families

The inference family establishes what may legitimately be claimed. It is epistemic: it audits the match between the operator a claim asserts and the operator its evidence supports. The control family governs what is done with the result. It is regulative: it takes the structural grounding of a candidate output as a perception, compares it to a reference, and attenuates the output accordingly. The inference operators detect the error; the control operators act on it. The two families meet at the attenuation operator O6, where a verdict about warrant becomes a force on the emitted distribution.

3.3 Categories and operators

The Abstract Category at which a claim sits shapes which operator it is apt to assert, though it does not fix it. A claim positioned at C5 (transformation / operation) is already about an operation and therefore reaches naturally for O2 or O3 — it speaks of something being brought about. A claim at C2 (process) describes an unfolding and is most often O1, unless it names a mechanism that licenses an ascent. Claims at C1 (entity), C3 (relation), and C4 (state) are typically descriptive and sit at O1; they cross into the causal operators only when they assert that an entity, relation, or state produces an effect. A claim at C6 (schema) is meta-level and, like an objective-necessary quadrant, usually asserts no empirical causal operation at all. These are tendencies the engine uses as priors on the asserted operator, to be confirmed against the claim's actual vocabulary — not substitutes for reading it.

Part II

The Inference Operators — Pearl's Ladder

4.01 — Conditioning

The first inference operator is conditioning: the operator of association, the first rung of the causal hierarchy. It is the operation of seeing — of reading one quantity from another within a fixed population.

4.1 Formal definition

For random variables X and Y , conditioning produces the conditional distribution

$$P(Y | X) = P(X, Y) / P(X), \text{ for } P(X) > 0.$$

This is a statement about co-occurrence. It answers the question: given that X is observed to take a value, how is Y distributed? It is computed entirely from the joint distribution and requires no causal assumption whatsoever.

4.2 What conditioning licenses

Conditioning licenses prediction and diagnosis under stable conditions. If the data-generating process does not change, observing X improves an estimate of Y exactly to the degree that they are statistically dependent. Conditioning is the proper operator for forecasting, pattern recognition, and the reading of symptoms — for any task in which the system is a passive observer of a population it does not disturb.

4.3 What conditioning does not license

Conditioning does not license any claim about the effect of acting. Reweighting a population by the observed value of X is not the same as setting X to that value. The distinction is the entire content of the next two operators. A system that uses O1 evidence to make an O2 or O3 claim has committed Unsupported Causality, the failure of Chapter 7. At HNS-144, a claim of this kind typically sits in Q3 (objective, contingent) and at category C2 or C5; it is precisely there that the engine must check that the asserted operator does not exceed O1.

4.4 Properties of the conditional distribution

Three properties are used throughout. The chain rule factorizes a joint distribution, $P(X, Y) = P(Y | X) P(X)$. Bayes' rule inverts a conditional, $P(X | Y) = P(Y | X) P(X) / P(Y)$. Marginalization recovers a lower-dimensional distribution by summing out a variable, $P(Y) = \sum_x P(Y | X = x) P(X = x)$. All three operate within a single joint distribution and introduce no causal content; they are bookkeeping on observation.

4.5 Threats at the first rung

Two structures make association an unreliable proxy for effect. A confounder is a common cause of X and Y ; conditioning on X then mixes the genuine path $X \rightarrow Y$ with the spurious path $X \leftarrow \text{confounder} \rightarrow Y$. A collider is a common effect of X and Y ; conditioning on it, or sampling on a variable it determines, induces a dependence where none existed — the mechanism of selection bias. Both are invisible to O1, which sees only the joint distribution, and both are exactly what the causal model of O2 is needed to expose. A system that treats a first-rung association as if these structures were absent has, in effect, asserted a causal model it never stated.

4.6 Measures of association

Conditioning is agnostic about how dependence is quantified. Correlation captures linear co-variation; mutual information captures dependence of any form, vanishing exactly when X and Y are independent; a conditional-independence test asks whether a dependence survives once a third variable is held fixed. All three are O1 instruments: they describe the joint distribution and license prediction, never effect. The engine notes which instrument an O1 claim rests on only to confirm that the claim stays within Rung 1; the choice among them never raises the rung.

4.7 A positioned O1 claim (non-normative)

Take “users who enable notifications open the app more often.” Positioned, it is an institutional behaviour read as a process, ($L6 \times C2$); observed, it is objective and contingent (Q3). Its asserted operator is O1 — an association. The claim is well formed and passes: it asserts exactly what a correlation supports. The engine logs it as O1 with O1 support, verdict pass, and would intervene only if a later step

rephrased it as “enabling notifications increases engagement,” an O2 ascent that would require its own warrant.

5.02 — Intervention (the do-operator)

The second inference operator is intervention. It is the operation of doing, the second rung of the hierarchy, and it is the operator most often invoked, often without warrant, whenever a system speaks of causes, effects, levers, or consequences.

5.1 The do-operator

The interventional distribution $P(Y \mid \text{do}(X = x))$ is the distribution of Y that results from setting X to x by external intervention, rather than from observing X to equal x . In the language of structural causal models, the intervention performs graph surgery: it removes the equation that ordinarily determines X , severing the arrows into X , and replaces it with the constant x . What flows downstream from X is preserved; what would have explained X is cut away.

5.2 Why intervention differs from conditioning

In general,

$$P(Y \mid \text{do}(X)) \neq P(Y \mid X).$$

The inequality is the signature of confounding. When a common cause influences both X and Y , the observed association $P(Y \mid X)$ mixes the influence of X on Y with the influence of the confounder on both, whereas $P(Y \mid \text{do}(X))$ isolates the former. The classic consequence is that a strong observed correlation can correspond to no interventional effect at all, and that the sign of an effect can even reverse between the two (Simpson's paradox is the discrete instance).

5.3 Identification

Intervention can sometimes be computed from observational data, but only with a causal model. If a set of variables Z satisfies the back-door criterion relative to (X, Y) — blocking every spurious path and containing no descendant of X — then the interventional distribution is identified by the adjustment formula:

$$P(Y \mid \text{do}(X)) = \sum_k P(Y \mid X, Z = z) \cdot P(Z = z).$$

Where no admissible back-door set exists, the front-door criterion or

the full do-calculus may still identify the effect; and where the graph licenses none of these, the effect is simply not identifiable from observation, and only experiment will settle it. The decisive point for the engine is structural: Rung 2 always requires a causal model — a set of assumptions about the graph — over and above the data. Evidence at Rung 1 never suffices on its own.

5.4 Role in the engine

Operator O2 is the proper operator for any claim about effects, policies, or mechanisms — for category C5 (transformation/operation) claims especially. The engine’s task is to confirm that a claim asserting O2 is accompanied by what O2 requires: an experiment, or an observational identification with its assumptions made explicit. A bare correlation offered in support of an O2 claim is the canonical Unsupported Causality, and is to be flagged.

5.5 The three rules of do-calculus

When an effect must be identified from observation together with a graph, the do-calculus supplies three rules that rewrite expressions containing the do-operator until none remains. In words: Rule 1 inserts or deletes an observation that is irrelevant to the outcome given the rest; Rule 2 exchanges an action for an observation — $\text{do}(X)$ for X — when the relevant back-door paths are blocked; and Rule 3 inserts or deletes an action that has no effect on the outcome along open paths. An effect is identifiable precisely when repeated application removes every do-operator, leaving a formula in observational quantities. The calculus is complete: if the three rules cannot eliminate the do-operator, no observational formula for the effect exists, and only experiment will settle it.

5.6 The front-door criterion

Where no admissible back-door set exists — because the confounder is unobserved — an effect may still be identified through a mediator M that lies wholly on the path from X to Y , is unconfounded with Y given X , and is shielded from the unobserved confounder. The front-door formula chains two adjustments:

$$P(Y \mid \text{do}(X)) = \sum_m P(M = m \mid X) \sum_{x'} P(Y \mid M = m, X = x') P(X = x').$$

The lesson for the engine is unchanged: identification is possible, but only relative to an explicit causal structure that the claim must declare.

5.7 A numerical instance: Simpson’s paradox

The gap between O1 and O2 is not a technicality; it can reverse a conclusion. In the figures below a treatment recovers a higher fraction of patients in each severity subgroup, yet a lower fraction in the pooled total, because the severe cases — harder to recover — were treated more often.

Group	Treated (recovered)	Untreated (recovered)
Mild cases	81 / 87 (93%)	234 / 270 (87%)
Severe cases	192 / 263 (73%)	55 / 80 (69%)
Pooled	273 / 350 (78%)	289 / 350 (83%)

Reading the pooled association $P(\text{recovery} \mid \text{treated}) < P(\text{recovery} \mid \text{untreated})$ as an effect — asserting O2 on O1 — would recommend withholding a treatment that in fact helps within every subgroup. The adjusted quantity $P(\text{recovery} \mid \text{do}(\text{treated}))$, which conditions on severity before pooling, restores the correct ordering. This is the practical face of Unsupported Causality.

5.8 Randomization as the gold standard

The cleanest warrant for an O2 claim is a randomized intervention. When X is assigned by a mechanism independent of every other variable, the arrows into X are severed by construction, so observation and intervention coincide within the experiment, $P(Y \mid \text{do}(X)) = P(Y \mid X)$. Randomization is therefore not one identification strategy among others but the case in which identification is automatic. An O2 claim backed by a randomized trial is licensed without further graphical argument; the back-door and front-door routes exist precisely for the common case in which randomization was not possible.

5.9 A back-door adjustment (worked, non-normative)

Suppose a claim asserts that a training programme raises wages (O2). Workers self-select into training by prior skill, which also raises wages, so prior skill is a back-door confounder. If prior skill is

measured and blocks the only spurious path, the adjustment formula identifies the effect:

$$P(\text{wage} \mid \text{do}(\text{train})) = \sum P(\text{wage} \mid \text{train}, \text{skill}) \cdot P(\text{skill})$$

The engine does not compute this sum; it checks that the claim names an adjustment set and that the set is defensible — that it blocks the spurious path and contains no descendant of training. A claim that adjusts for a post-training variable, a mediator or a consequence, fails the criterion and is flagged, because conditioning on a descendant of the cause distorts the very path the adjustment was meant to close.

5.10 A graph and an adjustment (worked, non-normative)

Let the causal structure be $Z \rightarrow X$, $Z \rightarrow Y$, and $X \rightarrow Y$, with Z observed. Z is a common cause of treatment X and outcome Y , and the back-door path $X \leftarrow Z \rightarrow Y$ is the only spurious route. The set $\{Z\}$ blocks it and contains no descendant of X , so it satisfies the back-door criterion and the effect is identified by adjustment:

$$P(Y \mid \text{do}(X)) = \sum_k P(Y \mid X, Z = z) \cdot P(Z = z)$$

With toy numbers — $P(Z = 1) = 0.5$; in the stratum $Z = 0$, $P(Y = 1 \mid X = 1) = 0.6$ and $P(Y = 1 \mid X = 0) = 0.5$; in the stratum $Z = 1$, $P(Y = 1 \mid X = 1) = 0.8$ and $P(Y = 1 \mid X = 0) = 0.7$ — the adjusted effect is $(0.6 - 0.5)(0.5) + (0.8 - 0.7)(0.5) = 0.10$, a ten-point causal increase. The unadjusted contrast is larger, because treated cases are over-represented in the high-outcome stratum $Z = 1$; reading that inflated contrast as the effect is the confounding sub-form of Unsupported Causality. The engine's check is satisfied here because the claim named $\{Z\}$ and $\{Z\}$ is a valid back-door set; the arithmetic it leaves to the analyst.

6.03 — Counterfactual

The third inference operator is counterfactual: the operation of imagining, the third and highest rung. It asks what would have been, in a specific case, had something been otherwise.

6.1 Formal content

A counterfactual quantity is written

$$P(Y_x \mid X', Y'),$$

read as the probability that Y would have taken the value Y_x had X been set to x , given that in fact X took the value X' and Y took the value Y' . Its evaluation requires a fully specified structural causal model — the functions relating each variable to its parents, together with a distribution over the exogenous noise — and proceeds in three steps: abduction (update the noise distribution on the observed evidence), action (apply the intervention to the model), and prediction (read off the resulting distribution of Y).

6.2 The hierarchy does not collapse

Pearl's Causal Hierarchy Theorem establishes that the three rungs are genuinely distinct: with probability one, interventional facts are not determined by observational facts alone, and counterfactual facts are not determined by interventional facts alone. The ladder cannot be climbed by cleverness with lower-rung data; each ascent requires a stronger model. This is why the engine treats the rung asserted by a claim as load-bearing rather than cosmetic: a counterfactual asserts the most and so must carry the most.

6.3 Role in the engine

Operator 03 is the operator of attribution and responsibility — of but-for reasoning, of “the outcome was due to” and “had we not, then.” Such claims are common and rhetorically powerful, and they are the most easily unsupported of all, because they require a structural model the speaker rarely possesses. A counterfactual offered with only interventional or, worse, associational support is Unsupported Causality at its most acute.

6.4 Structural causal models

A structural causal model is a tuple $M = \langle U, V, F, P(U) \rangle$: a set U of exogenous background variables with a distribution $P(U)$; a set V of endogenous variables; and a set F of structural functions, one per endogenous variable, expressing each as a function of its parents in V and of some exogenous variables. The model induces both a directed graph and a joint distribution. An intervention acts on F , replacing a function by a constant; a counterfactual acts on $P(U)$, conditioning on evidence before intervening. The model, not the data, is the carrier of causal meaning — which is why a claim at O2 or O3 is a claim about a model.

6.5 The three-step evaluation

A counterfactual $P(Y_x \mid X', Y')$ is evaluated on M in three steps. Abduction: update $P(U)$ to $P(U \mid X', Y')$, absorbing what actually happened into the background. Action: form the modified model in which the equation for X is replaced by $X = x$. Prediction: compute the distribution of Y in the modified model under the updated background. The procedure is deterministic given M ; its whole difficulty lies in possessing M , which is why counterfactuals are the most demanding operator the engine recognizes.

6.6 Probabilities of causation

Counterfactual analysis makes precise the everyday notions of necessary and sufficient cause. The probability of necessity, PN — the probability that Y would not have occurred had X not, given that both did — measures but-for causation. The probability of sufficiency, PS , measures whether X would bring about Y in a case where neither occurred. Both are third-rung quantities, frequently only bounded rather than point-identified from data, and asserting them — as legal, medical, and explanatory language constantly does — commits the speaker to a structural model whether or not one is acknowledged.

6.7 A probability of necessity (worked, non-normative)

Consider “the alarm is why the fire was contained” — an O3 attribution. The relevant quantity is the probability of necessity: had the alarm not sounded, would the fire have spread, given that it did

sound and was contained? From observational and even experimental data this is generally only bounded, not point-identified; monotonicity assumptions tighten the bounds but must themselves be argued. The engine therefore treats a bare necessity claim as O3-asserted with, at best, O2 support, and attenuates the individual-attribution force while leaving any licensed population effect — “alarms reduce spread on average” — intact. The lesson is general: but-for language is counterfactual and inherits the counterfactual’s heavy warrant.

6.8 A counterfactual on a toy model (worked, non-normative)

Take a linear structural model with exogenous noise U_Y and a binary treatment X , in which the outcome is generated by $Y = 2X + U_Y$. Observe one unit with $X' = 1$ and $Y' = 5$. The counterfactual “had X been 0, what would Y have been?” is evaluated in the three steps of Chapter 6.5.

Abduction: from $Y' = 2 \cdot 1 + U_Y = 5$ recover $U_Y = 3$ for this unit. Action: set $X = 0$, giving $Y = 2 \cdot 0 + U_Y$. Prediction: $Y = 0 + 3 = 3$. So for this unit “had X been 0, Y would have been 3” is licensed by the model, and the individual effect of the change is $5 - 3 = 2$ — the model’s coefficient. The point is not the arithmetic but its precondition: the conclusion required the full model, the equation together with the recovered noise, which is exactly what an O3 claim asserts it possesses. Strip away the model and the same sentence is an O3 assertion with no O3 support — the counterfactual-without-a-model sub-form.

7. Unsupported Causality

Unsupported Causality is the characteristic failure of the analytical resolution and the principal target of the inference operators. This chapter defines it formally, gives its detection rule, and works examples. It also gathers the analytical failure types of the engine and maps them onto the inherited failure taxonomy.

7.1 Definition

Order the inference operators by rung: $O1 < O2 < O3$. Each claim carries two operator labels: the operator it asserts (the strength of the conclusion it draws) and the operator its evidence supports (the strength the available warrant can sustain). Unsupported Causality is the condition

$$\textit{asserted operator} > \textit{supported operator}.$$

That is, the claim climbs higher on the ladder than its evidence can carry it. The commonest instances are an O2 claim on O1 evidence (“X is correlated with Y, therefore X causes Y”) and an O3 claim on O2 or O1 evidence (“the effect was due to X,” offered without a structural model).

7.2 Detection rule

The detection rule is local and mechanical. For each claim: (1) determine the asserted operator from its modal and causal vocabulary — “causes,” “raises,” “if we do” signal O2; “would have,” “but for,” “was responsible for” signal O3. (2) Determine the supported operator from the evidence actually offered — a correlation supports O1; an experiment or an explicit identification supports O2; a structural model with abduction supports O3. (3) Flag the claim when the asserted operator outranks the supported operator, and either require the missing warrant or attenuate the output. The check does not assess whether the claim is true; it assesses whether the asserted operation is earned.

7.3 Worked examples

Case 1. Output: “Cities with more police have more crime, so police

cause crime.” The vocabulary asserts O2 (cause). The evidence — a cross-sectional correlation — supports only O1. Asserted O2 > supported O1: Unsupported Causality. The correction either retreats to O1 (“police presence and crime are positively associated across cities”) or supplies O2 warrant (an identification adjusting for the confounder, here the underlying crime rate that drives both).

Case 2. Output: “Had the firm not advertised, it would have failed.” The vocabulary asserts O3 (counterfactual attribution). No structural model is offered; at most, advertised firms are observed to survive more often — O1 evidence. Asserted O3 > supported O1: a severe instance. The correction states the modest claim the evidence supports (“advertising is associated with survival”) or builds the model the counterfactual requires.

Case 3 (a correct ascent). Output: “In the randomized trial, the drug lowered mortality.” The vocabulary asserts O2. The evidence is a randomized experiment, which supports O2 directly. Asserted equals supported: the claim passes. The engine’s job is not to doubt causation everywhere, but to require that the rung be earned.

7.4 The analytical failure types and their mapping

The analytical engine recognizes a small family of failures, inherited in substance from the analytical specification and re-expressed in operator terms. They map onto the foundational and observational failure taxonomy as follows.

Analytical failure	Operator-level reading	Inherited type
Causal Misidentification	asserted rung exceeds supported rung	Unsupported Causality
Layer Drift	operator ranges over the wrong Natural Layer	Layer Jump
Category Misalignment	operator applied under the wrong Category	Category Ambiguity
Depth Confusion	rung treated as interchangeable with a lower one	modal overreach (HNS-144)
Resolution Collapse	configuration under-specified; operator unassigned	loss of auditability

Causal Misidentification is Unsupported Causality by another name

and is the engine’s central concern. The others are localizations of where, in the configuration, the analysis has gone wrong: in the layer, in the category, in the rung, or in the completeness of the configuration itself.

7.5 The operator order is strict

The inference operators form a strict order under the relation requires at least as strong a model as: $O1 < O2 < O3$. The order is strict by the Causal Hierarchy Theorem — a higher operator is never reducible to a lower one — so the gap that an Unsupported-Causality violation opens cannot be closed by accumulating more evidence of the lower kind. It can be closed only by a stronger model or an experiment. This is precisely why the engine’s remedy is to demand the missing warrant or to attenuate, and never to manufacture the higher rung from lower evidence: no quantity of correlation becomes an intervention, and no set of interventions becomes a counterfactual, by aggregation alone.

7.6 Vocabulary cues for the asserted operator

The asserted operator is read from a claim’s causal and modal vocabulary. The cues below are heuristic, not definitional, but they localize the rung a claim reaches for.

Cue vocabulary	Asserted operator
“is associated with,” “tends to,” “predicts,” “correlates”	O1 — conditioning
“causes,” “raises,” “leads to,” “if we do X then”	O2 — intervention
“would have,” “but for,” “was responsible for,” “due to”	O3 — counterfactual

7.7 Sub-forms of Unsupported Causality

Unsupported Causality is a single condition — asserted rung above supported rung — but it arises through a small number of recurring structures. Naming them helps a verifier localize the missing warrant.

- **Confounding.** A common cause of X and Y creates or inflates the association; the O2 claim needs an adjustment set that blocks the back-door path. Correction: adjust, or retreat to O1.

- **Collider / selection.** Conditioning on a common effect, or sampling on something it determines, manufactures an association; the remedy is to stop conditioning on the collider, not to find a stronger correlation.
- **Reverse causation.** The association is real but the arrow runs $Y \rightarrow X$; temporal or interventional evidence is needed to fix the direction.
- **Over-adjustment / mediation.** Adjusting for a variable on the causal path removes part of the very effect being claimed; correction is not to condition on mediators when estimating a total effect.
- **Counterfactual without a model.** An O3 attribution is asserted with no structural model to support abduction; correction is to build the model or down-rank to the supported interventional or associational claim.
- **Extrapolation beyond support.** An identified effect is transported to a population or regime where its identifying assumptions do not hold; the warrant lapses outside the support of the data.

Each sub-form is the same violation localized to a different structure, and each admits the same two remedies the engine ever offers: supply the missing warrant, or attenuate the over-claim.

7.8 Modality and the licensed operator

The quadrant a claim occupies at HNS-144 shapes which operator it may legitimately assert. An objective-contingent claim (Q3) is the natural home of O1–O3: it is about how the world in fact is, and so may ascend the causal ladder if it earns the warrant. An objective-necessary claim (Q4) — a definition or a theorem — asserts no causal operation at all, and reading it as O2 is a category error, not merely an over-claim. A subjective claim (Q1, Q2) asserts a preference or a commitment; a causal operator applied to it crosses the subjective/objective boundary before the question of rung even arises. The engine therefore uses the quadrant as a gate: it asks first whether an inference operator is appropriate to the modality at all, and only then whether the asserted rung is supported.

Quadrant	Typically licensed operators	Caution
Q1 Subj × Cont	O1 within the subjective frame	causal operators cross the domain boundary
Q2 Subj × Nec	none inferential (constitutive)	not an empirical causal claim
Q3 Obj × Cont	O1, O2, O3 (with warrant)	the principal site of Unsupported Causality
Q4 Obj × Nec	none inferential (definitional)	reading as O2 is a category error

Part III

The Control Operators — Perceptual Control Theory

8.04 — Reference Signal

The control family begins with the reference. Perceptual Control Theory holds that an organism does not control its actions or its outputs directly; it controls its perceptions, acting so as to bring a perceived variable to a desired value. That desired value is the reference signal.

8.1 The reference invariant

In the control loop, the reference r is the specification of the intended value of a controlled variable. It is set from outside the loop — by the goal, the standard, or in the present setting the verification specification — and is held invariant while the loop runs. In HNS, the reference is the grounding target: the structural state a candidate output is required to match — fully positioned, with every asserted operator supported, and compatible with the grounding stack of the social-meta layer.

The reference is not itself derived from the output; that is what makes it a reference. Its role in the engine is to give the comparator something to measure against, so that the gap between what an output is and what it should be becomes an explicit, signed quantity rather than an implicit judgement.

8.2 Setting the reference

The reference is a policy, and naming it explicitly is part of what makes the engine auditable. A strict reference demands near-complete grounding before an output passes; a lenient reference admits weakly grounded output and relies on attenuation to scale it. Neither is correct in the abstract: a medical or legal deployment sets a high reference and accepts more refusals, while a brainstorming aid sets a low one and accepts more speculation. What the specification requires is not a particular value but that the value be declared and logged, so that the standard against which outputs were judged can be reconstructed.

9.05 — Comparator and Error Detection

The second control operator is the comparator: the element that computes the gap between reference and perception and drives the loop to close it.

9.1 The control error

The comparator forms the control error

$$e = r - p,$$

the difference between the reference r and the current perception p . The loop is a negative-feedback loop: action is generated so as to reduce the magnitude of e , returning the controlled perception toward the reference. In HNS, the perception p is the perceived structural state of a candidate output — the degree to which it is positioned and its operators supported — and the error e measures how far that falls short of the grounding target. A large error is a structurally ungrounded output; a zero error is a fully grounded one.

9.2 Relation to predictive coding

Note. Friston's free-energy principle and the predictive-coding scheme of Rao and Ballard minimize a prediction error, the difference between sensory input and a top-down prediction. This is a structural analogue of the comparator and may be used as an alternative formalization in active-inference implementations. It is not identical: Perceptual Control Theory acts to keep a perception at an externally set reference, whereas predictive coding acts to minimize surprise relative to a generative model. The two coincide only when the reference is read as the prediction; the engine treats predictive coding as an admissible implementation of O5, not as its definition.

What the comparator contributes to the engine is a signed, quantitative measure of structural shortfall. That measure is exactly what the final operator needs: it is the input from which the grounding score, and hence the attenuation, is computed.

9.3 The feedback law

In discrete time the loop drives the controlled perception toward the reference. Writing $p(t)$ for the perception at step t , r for the reference, and k for the loop gain, a first-order controller makes the error $e(t) =$

$r - p(t)$ evolve as

$$e(t+1) = (1 - k) \cdot e(t).$$

The error decays to zero, and the loop is stable, when $|1 - k| < 1$, that is $0 < k < 2$; higher gain converges faster but risks overshoot. In the engine the gain is not a free dynamical parameter but a design choice governing how sharply a structural shortfall is corrected before emission.

9.4 Hierarchies of control

Perceptual Control Theory composes such loops into a hierarchy in which the output of a higher loop sets the reference of a lower one. The engine uses the same idea: the social-meta grounding stack of Volume 4 supplies references that the structural comparator then controls toward. The present volume specifies only the single loop whose actuator is output attenuation.

9.5 Convergence of the loop (worked, non-normative)

Take a reference $r = 1$ (full grounding) and an initial perception $p(0) = 0.4$, with loop gain $k = 0.5$. The error $e(0) = 0.6$ decays as $e(t+1) = (1 - k) e(t) = 0.5 e(t)$: 0.6, 0.3, 0.15, ... toward zero. With $k = 1$ the error is annihilated in a single step; with $k = 1.5$ it oscillates while shrinking (0.6, -0.3, 0.15, ...); with $k \geq 2$ it diverges. In the engine this dynamic is not literally iterated against a fixed output — each output is judged once — but the same arithmetic fixes how aggressively a measured shortfall is converted into attenuation, and the stability band $0 < k < 2$ marks the regime in which correction is well behaved.

10.06 — Output Attenuation

The third control operator is the one original to HNS. It is the actuator of the loop: the element that converts the verdict of the inference and comparator operators into a force on the emitted output. Where the previous operators detect and measure, O6 acts.

10.1 The grounding score

Let $s \in [0, 1]$ denote the structural grounding score of a candidate output: a monotone function of the control error, equal to one when the error is zero and decreasing as the error grows. Informally, s answers: to what degree is this output positioned, operator-supported, and grounded? It is one when the asserted operators are fully earned and the perception matches the reference; it is zero when the output is structurally ungrounded — for example, when it asserts a causal conclusion with no warrant of the required rung.

10.2 The attenuation map

The attenuation operator applies a map g to the grounding score and scales the raw output by the result:

$$\text{output} = g(s) \cdot \text{raw_output}, \text{ with } g : [0,1] \rightarrow [0,1], g(0) = 0, g \text{ monotone non-decreasing.}$$

Here `raw_output` denotes the pre-decode generative signal — the probability or logit vector before token selection — so that attenuation shapes generation rather than filtering text after the fact. The conditions on g are deliberately minimal: it must vanish at zero grounding, it must never penalize better grounding, and it must remain a fraction. Its exact shape — linear, thresholded, or sigmoidal — is an engineering parameter, tuned to trade fluency against caution, and is not fixed by the specification.

10.3 Zero-Incidence Attenuation

The defining property of O6 is its behaviour at the boundary.

Proposition (Zero-Incidence Attenuation). *If $s = 0$ then $\text{output} = 0$.*

The proof is immediate: by definition $g(0) = 0$, so $\text{output} = g(0) \cdot$

raw_output = 0 regardless of the raw output. The content of the proposition is not its difficulty but its guarantee: a fully ungrounded output is not attenuated to something small but emitted anyway — it is set to zero. The system cannot, by construction, emit content that scores zero on structural grounding. This is the sense in which HNS aims at verification rather than mere improvement: the floor is structural, enforced by the actuator, not statistical.

10.4 Placement and the meeting of the families

Attenuation is the point at which the inference and control families meet. The inference operators decide whether an asserted operation is supported; the comparator turns the verdict into an error; the grounding score turns the error into a number in the unit interval; and O_6 turns that number into a multiplicative force on the generative distribution. A failure detected at Rung 2 — an unsupported causal claim — lowers s , and a low s attenuates the very tokens that would have expressed the unsupported claim. The audit and the action are one mechanism.

10.5 Shapes of the attenuation map

Three shapes of g are useful. The linear map $g(s) = s$ attenuates in proportion to shortfall. The threshold map, $g(s) = 0$ for $s < \tau$ and $g(s) = 1$ for $s \geq \tau$, enforces a hard floor at a chosen $\tau \in [0, 1]$, emitting only outputs above the threshold. The power or sigmoid map, $g(s) = s^\gamma$ with $\gamma \geq 1$ or a logistic curve centred at τ , interpolates between the two — sharpening near the threshold while remaining smooth. All satisfy $g(0) = 0$ and monotonicity; the choice trades fluency against caution and is fixed per deployment, not by the specification.

10.6 Composing the grounding score

The grounding score aggregates per-operator verdicts. Let $a_m \in [0, 1]$ be the support margin of operator O_m for a claim — one when the asserted operator is fully earned, zero when it is wholly unwarranted. A conservative composition takes the minimum, $s = \min_m a_m$, so a single unsupported operator drives the score down; a graded composition takes a weighted product, $s = \prod_m a_m^{w_m}$ with $\sum w_m = 1$. Both are monotone non-decreasing in each a_m , so improving the

support of any operator never lowers the score — a property the attenuated output inherits through the monotonicity of g .

10.7 Acting before the decode

Acting before token selection means acting on the distribution. If the raw model assigns logits z over the vocabulary, attenuation by $g(s)$ can be realized as a contraction of probability mass toward a grounded baseline as s falls — a temperature-like reweighting that down-weights ungrounded continuations relative to grounded ones. In the limiting case $s = 0$ the contraction is total: by Zero-Incidence Attenuation no ungrounded continuation retains mass, and the ungrounded output is not produced. The mechanism is generative, not a post-hoc filter on finished text.

10.8 A worked attenuation

Suppose a claim asserts O2 but offers only O1 support, so its causal support margin is $a = 0.2$ while its other operators are fully supported. Under the conservative composition $s = \min = 0.2$; under the linear map $g(s) = 0.2$, so the causal phrasing is carried at one-fifth strength and, in the pre-decode reweighting, falls below competing grounded continuations. Supplying an identification raises a toward one, s toward one, and the attenuation lifts. The numbers are illustrative; the behaviour — monotone, bounded, and zero at the floor — is guaranteed.

10.9 The attenuation map tabulated

The three admissible families behave as follows on representative grounding scores: the linear map $g(s) = s$, the threshold map at $\tau = 0.6$, and the power map $g(s) = s^2$. All vanish at $s = 0$ and reach 1 at $s = 1$; they differ in the middle, where the power map is most forgiving of moderate grounding and the threshold map least.

s	$g = s$ (linear)	threshold $\tau = 0.6$	$g = s^2$ (power)
0.0	0.00	0	0.00
0.2	0.20	0	0.04
0.4	0.40	0	0.16
0.6	0.60	1	0.36

s	g = s (linear)	threshold $\tau = 0.6$	g = s^2 (power)
0.8	0.80	1	0.64
1.0	1.00	1	1.00

The choice among them is the single most consequential tuning decision in a deployment, and is properly made against a measured error profile rather than asserted in advance; what the specification fixes is only the three axioms every admissible map shares.

Part IV

Integration and Status

11. The Generative Principle and Operator Accuracy

This chapter states how an 864 configuration is constructed and what it means to apply the engine correctly.

11.1 The generative principle

An analytical configuration is generated in three movements. First, the claim is positioned in HNS-36 by its causal layer and explanatory category. Second, it is observed in HNS-144 by its modal quadrant. Third, the operator is applied: the claim is read as performing one of the six operations, and the engine checks the operation against its warrant. The full space of 864 configurations is the product of these choices, and every claim that the engine processes occupies exactly one configuration at a time — the analytical analogue of the single-assignment rule of the foundational coordinate.

11.2 Operator accuracy

Operator accuracy is the requirement that the correct operator be applied to the correct claim. Two errors are symmetric. To apply an operator above the supported rung is to over-claim — Unsupported Causality, the dangerous direction, in which a system asserts more than it knows. To apply an operator below the supported rung is to under-claim — a waste of available warrant, in which a system declines to draw a conclusion it has earned. The engine is tuned chiefly against the first, because over-claiming is what makes a fluent system unreliable; but a complete account names both, since a verifier that never permits a causal claim is as useless as one that permits every causal claim.

Operator accuracy is, like every other rule of HNS, an external and auditable check. The operator asserted by a claim and the operator supported by its evidence are both recorded; their comparison is the verdict; and the verdict is logged. Nothing in the procedure requires access to the internals of the generating model.

11.3 Operator trajectories

A chain of reasoning is analysed not claim by claim alone but as an operator trajectory: the ordered sequence of operators its steps assert, read alongside the cell trajectory of the foundational volume. The trajectory exposes a failure invisible at the level of single claims — an illegitimate ascent, in which a chain accumulates only associational support yet concludes with an interventional or counterfactual claim. The rule is that the operator asserted by a conclusion may not outrank the strongest operator its premises jointly support, and that support does not accumulate across rungs: any number of O1 links never compose into an O2 conclusion.

Worked trajectory 1 — an illegitimate ascent

Premise (O1): “regions with more clinics report better health outcomes.” Premise (O1): “wealthier regions have more clinics.” Conclusion (O2): “building clinics improves outcomes.” Both premises are associational; the conclusion is interventional. The trajectory O1, O1 → O2 is an ascent without warrant: the shared cause, wealth, is exactly the unblocked back-door path. The engine flags the conclusion and either demands an identification or attenuates it, leaving the premises intact.

Worked trajectory 2 — a licensed chain

Premise (O2, randomized): “in trials, the vaccine lowers infection.” Premise (O1): “this population resembles the trial population on measured covariates.” Conclusion (O2, transported): “the vaccine will lower infection here.” The conclusion stays at O2 and rests on an O2 premise plus an explicit transport assumption; the trajectory does not ascend beyond its support. The engine passes it while logging the transport assumption as the claim’s load-bearing premise, so that a later challenge can be aimed exactly at it.

Trajectory analysis is the analytical analogue of the layer-jump check. There, the offence is a non-adjacent jump across Natural Layers; here, it is a non-adjacent jump up the causal ladder. Both are detected mechanically from the sequence of coordinates, without recourse to whether the conclusion happens to be true.

11.4 The single-operator rule and its limits

Two requirements complete the account of correct application. First, single assignment: an atomic claim performs exactly one operation and so occupies exactly one operator at a time; a sentence that performs several is decomposed before analysis, as in the composite cases of the casebook. Second, honest extraction: the asserted operator is read from the claim's vocabulary and structure, and where the vocabulary is genuinely ambiguous — “affects,” “drives,” “shapes” can be read as O1 or O2 — the engine records the ambiguity rather than resolving it silently, and treats the claim conservatively, holding it to the higher rung's warrant unless the lower reading is explicitly marked. The limit of the method lies exactly here: operator extraction from natural language is itself a judgement, and the inter-rater reliability of that judgement is the empirical quantity on which the engine's practical value most depends.

12. Architecture and Runtime Placement

The analytical engine is the finest resolution of the structural axis, Axis 2 of the Multi-Axis Verification account. This chapter situates it among the other axes and describes its runtime form.

12.1 Position among the three axes

Axis 1 (generative) proposes; it is the model's native fluency. Axis 2 (structural/causal), of which HNS-864 is the analytical layer, constrains: it positions a proposal, observes its modal stance, and audits the operation it asserts. Axis 3 (grounded/executive), the social-meta layer of Volume 4, checks the proposal against context from the body to the institution. Reliability is the conjunction of the three. HNS-864 contributes the causal-structural condition and is silent, by design, on raw probability and on world-grounding, which belong to Axes 1 and 3.

12.2 Runtime form

At runtime the engine sits between the generative distribution and token selection. A candidate output is positioned and observed; its asserted operators are extracted and compared to their support; the comparator forms the control error; the grounding score is computed; and the attenuation map scales the pre-decode distribution. Every step writes a machine-readable record — the configuration, the asserted and supported operators, the score, and the attenuation applied — so that the verdict is reconstructable after the fact. The independent verification architecture of the Volume 5 specifies that log format and the hardware-level stop that bounds it; the present volume specifies only what is computed, not how it is deployed.

12.3 Where Unsupported Causality is caught

In the runtime loop, Unsupported Causality is caught at the comparison step and acted upon at the attenuation step. A claim whose asserted rung exceeds its supported rung lowers the grounding score; the lowered score attenuates the tokens that would have expressed the over-claim. The detection is epistemic and the response is regulative, but they are realized as a single pass over the

candidate output.

12.4 The log record

Every pass writes a machine-readable record so that the verdict is reconstructable. The record carries, at minimum: the claim as positioned, written as its configuration ($L \times C \times Q \cdot O$); the asserted operator and the supported operator; the grounding score s ; the attenuation $g(s)$ applied; and, where a flag was raised, its sub-form and the missing warrant. Because every field is external to the generating model — derived from the claim's coordinate and its declared evidence, not from the model's internal states — the record can be audited by a third party who never sees the model. The format and its cryptographic commitments are specified in the external-verification volume; the present volume fixes only the fields the engine must emit.

12.5 Relation to the three axes, restated

It bears repeating where the engine's authority ends. Axis 1 decides how probable a continuation is; Axis 3 decides whether it is grounded in body, dialogue, organization, economy, governance, and the wider system. Axis 2, and the 864 engine within it, decides only whether the operation a claim performs is structurally warranted. A claim may be probable, structurally warranted, and still false because it is ungrounded in fact; conversely a true claim may be flagged here because, as stated, it asserts more than its declared evidence licenses. The engine is one of three conditions, not the whole of reliability.

12.6 What the engine does not do

The engine does not establish truth, does not estimate probability, and does not repair a claim. It detects a mismatch between asserted and supported operation and converts that mismatch into an attenuation. Repair — supplying the missing identification, building the missing model, or rephrasing to the supported rung — is the work of the generating system or its author, prompted by the flag. The engine's discipline is precisely that it does less than judge content: it judges licence, which is all that can be checked from outside the model and all that is needed to refuse an over-claim.

13. Intellectual Lineage and Proponents

The analytical engine recombines several established bodies of work. This chapter records the debts and, as in the foundational volume, is candid that the proponents did not design HNS.

13.1 The inference family

The inference operators are Judea Pearl's causal hierarchy in full: conditioning, the do-operator, and counterfactuals, together with the identification theory — back-door and front-door criteria and the do-calculus — and the Causal Hierarchy Theorem that keeps the rungs distinct. The structural-causal-model machinery underlying counterfactuals is also Pearl's, with antecedents in the structural equation tradition and the constraint-based causal discovery of Spirtes, Glymour, and Scheines.

13.2 The control family

The control operators rest on William Powers' Perceptual Control Theory, with its insight that behaviour is the control of perception, and on the classical negative-feedback control of Wiener's cybernetics. The comparator has a widely noted analogue in the predictive-coding scheme of Rao and Ballard and in Friston's free-energy principle, used here as an admissible alternative formalization. The orthogonality intuition that motivates crossing an external constraint plane against the generative axis draws on the information geometry of Amari.

13.3 What is HNS's own

The assignment of these six items as a single ordered axis, the multiplicative nesting that yields 864, the structural grounding score, and above all the output-attenuation operator with its zero-incidence guarantee are HNS's own. So is the decision to treat the whole apparatus as an external verifier rather than as a theory of cognition. Pearl's hierarchy is a theory of causal inference; Powers' theory is a theory of behaviour; neither was written to constrain a language model's output. The recombination, and the responsibility for it, are the author's.

Element	Proponent / source	Borrowed or HNS
01–03 inference ladder	Pearl	borrowed
Identification, do-calculus	Pearl; Spirtes et al.	borrowed
04–05 control loop	Powers; Wiener	borrowed
Comparator analogue	Rao & Ballard; Friston	borrowed (alternative)
Orthogonality intuition	Amari	borrowed
06 attenuation; 864 nesting; verifier use	Hara (HNS)	HNS

13.4 The proponents, in brief

Judea Pearl’s work supplies the entire inference family. His ladder of causation — seeing, doing, imagining — and the do-calculus that decides when a causal effect can be read from observation are the mathematical content of 01–03. HNS adopts them unchanged; what it adds is the decision to treat the rung a claim asserts as an auditable property of generated text.

William Powers’ Perceptual Control Theory supplies the control frame. Its central claim — that organisms control their perceptions, not their outputs, acting to hold a perceived variable at a reference — gives 04 and 05 their form. HNS reads the controlled perception as a grounding score and the reference as a verification standard.

The negative-feedback loop itself, and the requirement that a regulator command enough variety to counter the disturbances it faces, come from the cybernetics of Wiener and Ashby. They justify the loop’s structure and, in Ashby’s law of requisite variety, the intuition that a verifier must be at least as rich as the failures it screens. Amari’s information geometry motivates the picture of an external constraint surface crossing the generative manifold — the orthogonality that lets Axis 2 add information the generative axis lacks — while Friston’s free-energy principle and Rao and Ballard’s predictive coding furnish the comparator’s analogue.

None of these authors wrote for the purpose to which HNS puts their work, and none should be read as endorsing the framework. The borrowing is acknowledged so that the genuinely new commitments

— the sixfold axis, the 864 nesting, and the attenuation operator — remain visible as the author's own.

14. Empirical Status and Limitations

This final chapter states plainly what the specification establishes, what it proposes, and where it is exposed.

14.1 What is established

Established are the mathematical facts the engine inherits and its own well-definedness. Pearl's ladder, the adjustment and do-calculus identification results, and the Causal Hierarchy Theorem are settled mathematics. The negative-feedback control law and the elementary properties of the attenuation map — including the zero-incidence proposition — are likewise not in doubt. The 864 nesting is well-defined, its cardinality is exactly the product of its factors, and every step of the engine is externally auditable. These are the firm parts.

14.2 What is proposed

Proposed, and defensible but unproven, are the engineering commitments: that operator-support checking is a good operationalization of analytical reliability; that pre-decode attenuation is an effective place to act; and that the six operators are the right six. These are reasoned choices, argued in the preceding chapters, not consequences of the borrowed mathematics.

14.3 What remains conjecture

Conjecture, finally, are the quantitative claims about reliability — that the engine drives a system's hallucination rate toward a structural floor, that the attenuation materially reduces unsupported output in practice. These are empirical questions, reported on in the Origin Trilogy and open to further test; the present volume does not establish them.

14.4 Threats to the account

Several threats deserve naming. First, identifiability: Rung-2 conclusions drawn from observational data depend on a correct causal graph, and a wrong graph yields a confident wrong answer; the engine checks that an O2 claim offers identification, not that the identification is correct. Second, circularity in the grounding score: if

s is computed by the same framework whose output it gates, a self-consistent but mistaken analysis can score itself highly; the score's validity is therefore a matter for external calibration, not internal assertion. Third, operator-assignment reliability: deciding which operator a claim asserts can be a judgement, and inter-rater agreement on that assignment is a concrete, measurable target for validation. Fourth, the boundary with Axis 3: whether a defect is a missing causal warrant (Axis 2) or a false fact about the world (Axis 3) is not always sharp. Fifth, cross-lingual and cross-domain transfer of the operator vocabulary is untested. And the recurrence of six, here as elsewhere, is a chosen resolution. The specification is offered as a stable but revisable v1.0, not a closed result.

14.5 Two further exposures

Two further exposures deserve explicit mention. The first is operational: acting before the decode requires that the grounding score be computable within the latency budget of generation, and that the contraction of the probability vector be realizable without retraining the model. Both are engineering claims to be demonstrated, not assumed, and are the subject of the external-verification volume. The second is the elicitation of causal structure: every O2 and O3 verdict presupposes that the relevant causal graph can be stated, yet in open-domain language the graph is rarely given and must be inferred, which reintroduces judgement at exactly the point the engine aims to remove it. The honest position is that the engine converts a vague reliability question into a sharp, localized one — is this operator warranted, against this stated structure? — without claiming to answer it automatically in every case.

Appendix A — Normative Reference Tables

The tables below are normative. In case of conflict between prose and table, the table governs; in case of conflict with the Unified Structural Coordinate Table, that table governs.

A.1 Inference & Control Operators (the third axis)

Op.	Name	Lineage	Formal content
01	Conditioning	Statistics / Pearl	$P(Y X)$ — association (Rung 1)
02	Intervention / do	Pearl	$P(Y do(X))$ — intervention (Rung 2)
03	Counterfactual	Pearl	$P(Y_x X', Y')$ — counterfactual (Rung 3)
04	Reference Signal	Powers (PCT)	reference invariant r
05	Comparator / Error	Powers (PCT)	control error $e = r - p$
06	Output Attenuation	HNS Engineering	output = $g(s) \cdot raw$, $g(0)=0$, g monotone, $g(s) \in [0,1]$

A.2 The structural nesting

Layer	Construction	Cardinality
HNS-36	Natural Layers (6) × Abstract Categories (6)	36
HNS-144	HNS-36 × Onto-Modal Quadrants (4)	144
HNS-864	HNS-144 × Inference & Control Operators (6)	864

Note. The Social Meta Layers (SMS-6) of Volume 4 are Axis 3 and are orthogonal to this nesting: they ground and regulate output, they do not multiply the coordinate.

Appendix B — Configuration Template and Worked Configurations

A configuration is documented in the form below.

- Configuration: $(L_i \times C_j \times Q_k \cdot O_m)$.
- Asserted operator: the operation the claim performs.
- Supported operator: the strongest operation its evidence sustains.
- Verdict: pass if asserted $\leq$ supported, else flag (Unsupported Causality) and attenuate.

B.1 ($L6 \times C5 \times Q3 \cdot O2$) — a policy claim

Claim: “raising the rate reduces inflation.” Configuration: macro-social (L6), transformation (C5), objective-contingent (Q3), asserting intervention (O2). Supported operator: O2 only if an identification or experiment is offered; otherwise O1. Verdict: pass with identification; otherwise flag.

B.2 ($L3 \times C2 \times Q3 \cdot O1$) — a predictive claim

Claim: “users who hesitate here tend to abandon the form.” Configuration: cognitive (L3), process (C2), objective-contingent (Q3), asserting association (O1). Supported operator: O1 from the observed rates. Verdict: pass — the asserted operator does not exceed the support.

B.3 ($L4 \times C5 \times Q3 \cdot O3$) — an attribution

Claim: “had the review step not run, the error would have shipped.” Configuration: operational (L4), transformation (C5), objective-contingent (Q3), asserting counterfactual (O3). Supported operator: O3 only with a structural model; commonly only O1/O2 is available. Verdict: flag unless the model is supplied.

B.2 Further worked configurations

These configurations are non-normative and illustrate the template across the coordinate.

Claim: “smoking causes lung cancer.” Configuration: a physical-

substrate mechanism read as a transformation, objective and contingent, asserting intervention — ($L1 \times C5 \times Q3 \cdot O2$). Supported operator: O2, by extensive identification and experiment. Verdict: pass; the rung is earned.

Claim: “the market rose because investors felt optimistic.” Configuration: a macro-social process, objective and contingent, asserting intervention — ($L6 \times C2 \times Q3 \cdot O2$). Supported operator: at most O1, a co-movement of sentiment and prices. Verdict: flag; supply identification or retreat to association.

Claim: “by definition, a prime has exactly two divisors.” Configuration: a schema, objective and necessary — ($L4 \times C6 \times Q4$). No inference operator applies; the claim is definitional. Verdict: pass as Q4; a causal reading would be a category error.

Claim: “users prefer the blue button.” Configuration: a micro-social state, subjective and contingent — ($L5 \times C4 \times Q1 \cdot O1$). Supported operator: O1 within the subjective frame, a measured preference. Verdict: pass; an O2 reading (“blue causes preference”) would cross the domain boundary.

Claim: “had we shipped a day earlier, the outage would not have happened.” Configuration: an operational transformation, objective and contingent, asserting counterfactual — ($L4 \times C5 \times Q3 \cdot O3$). Supported operator: O3 only with a structural model of the failure; usually only O1 or O2 is available. Verdict: flag unless the model is supplied.

Claim: “the neuron fires when its input exceeds threshold.” Configuration: a cellular process, objective and contingent, asserting a mechanism — ($L2 \times C2 \times Q3 \cdot O2$). Supported operator: O2, a well-characterized mechanism. Verdict: pass.

Appendix C — Worked Examples

These examples are non-normative and illustrate the engine end to end.

C.1 Catching an over-claim

A system produces: “Teams that adopted the tool shipped faster, so the tool makes teams faster.” The engine positions the claim at (L6 × C5 × Q3), extracts an asserted O2, finds only O1 support (an adoption-speed correlation, with faster teams more likely to adopt), and flags Unsupported Causality. The grounding score falls; the attenuation map scales down the causal phrasing; the logged record shows asserted O2, supported O1, verdict flag.

C.2 Attenuation at the boundary

A candidate continuation asserts a confident counterfactual with no model and no data — a structurally ungrounded claim, $s = 0$. By Zero-Incidence Attenuation the output is set to zero: the continuation is not emitted at all, and the record notes the zero score. A neighbouring continuation that retreats to the supported associational claim scores near one and passes essentially unattenuated.

C.3 Under-claiming

Operator accuracy has a second, milder failure. A system answers “does the new compiler flag reduce bugs?” with “projects using it report fewer bugs” — an O1 statement — when a randomized rollout was in fact available that licenses the O2 claim “the flag reduces bugs.” Asserted O1, supported O2: the system declined a conclusion it had earned. The engine records the under-claim. It is not attenuated, since nothing ungrounded was emitted, but it is logged, because a verifier that silently wastes warrant is, over time, as unhelpful as one that over-claims.

C.4 A composite sentence

A single sentence often mixes rungs: “the popular new policy will obviously cut crime.” Decomposed, it is a subjective-contingent claim about popularity (Q1, O1), an objective-contingent prediction about crime (Q3, asserting O2), and a modal overreach in the word

“obviously.” The engine maps the parts separately, passes the popularity and the hedged prediction, and attenuates the causal-necessity overreach — accepting the defensible components while damping the one that is unearned, rather than passing or rejecting the sentence whole.

C.5 A full pass, end to end

A system is asked “why did revenue fall?” and answers “customers left because the price rose.” The engine decomposes the answer into an attrition claim and a causal attribution. It positions the attribution at ($L6 \times C5 \times Q3$), extracts asserted O2 from “because ... rose,” and looks for warrant: was price changed experimentally, or is there an identification adjusting for the confounders — seasonality, a competitor’s entry — that move both price and attrition? None is offered. The supported operator is O1. The support margin for the causal operator is low, the grounding score falls, and the linear map contracts the causal continuation below the grounded alternative, “revenue fell while price rose; the two moved together.” The log records the configuration, asserted O2, supported O1, the score, the attenuation, and the confounding sub-form. The author, seeing the flag, can supply the identification or accept the hedged statement; either way the over-claim does not ship unexamined.

Appendix D — Do-Calculus and a Worked End-to-End Pass

D.1 The three rules, restated

The do-calculus operates on a causal graph G and on graphs obtained by surgery: deleting arrows into a set (written with an overline) or out of it (written with an underline). Rule 1 (insertion / deletion of observations) holds when the outcome is independent of the observed set given the rest in the intervened graph. Rule 2 (action / observation exchange) replaces $\text{do}(X)$ by conditioning on X when X is independent of the outcome, in the graph with arrows out of X removed, given the adjustment set. Rule 3 (insertion / deletion of actions) removes $\text{do}(X)$ entirely when X is independent of the outcome in the graph with arrows into X removed, given the rest. Their repeated application is sound and complete for identification.

D.2 A full pass

Consider the output: “Students who used the tutoring app scored higher, so the app raises scores.” The engine positions the claim at $(L6 \times C5 \times Q3)$ — an institutional transformation, objective and contingent. It extracts the asserted operator $O2$ from “raises.” It asks for identification: is there a back-door set blocking the obvious confounder — that more motivated students both adopt the app and score higher? None is offered, and no front-door mediator is named. The supported operator is therefore $O1$. Since asserted $O2 \succ$ supported $O1$, the support margin for the causal operator is low, say $a = 0.2$; the grounding score $s = 0.2$; and the linear attenuation $g(s) = 0.2$ contracts the causal continuation’s mass below the grounded alternative — “using the app is associated with higher scores.” The log records: configuration $(L6 \times C5 \times Q3 \cdot O2)$; asserted $O2$; supported $O1$; $s = 0.2$; verdict flag; attenuation 0.2. A reader can reconstruct the verdict without access to the model that produced the sentence — the property the architecture exists to guarantee.

D.3 A multi-step reduction (sketch)

Where a single adjustment does not suffice, identification proceeds by repeated rewriting. Consider the front-door structure $X \rightarrow M \rightarrow Y$ with

an unobserved confounder between X and Y . The effect $P(Y \mid \text{do}(X))$ cannot be obtained by back-door adjustment, because the confounder is unobserved. The do-calculus reaches it in two moves: first identify $P(M \mid \text{do}(X)) = P(M \mid X)$, since no unblocked back-door path connects X and M ; then identify $P(Y \mid \text{do}(M))$ by adjusting for X , which blocks the back-door path from M to Y through the confounder. Composing the two and marginalizing over M yields the front-door formula of Chapter 5.6, now seen as the result of a two-step reduction rather than a standalone trick. The engine does not perform the reduction; it requires that a claim invoking the front-door route name the mediator and assert the shielding the route requires.

Appendix E — Notation Summary

The symbols used in this volume, collected for reference.

Symbol	Meaning
$(L_i \times C_i \times Q_k \cdot O_m)$	a full 864 analytical configuration
$P(Y \mid X)$	conditioning / association (01)
$P(Y \mid \text{do}(X))$	interventional distribution (02)
$P(Y_x \mid X', Y')$	counterfactual quantity (03)
$\text{do}(X = x)$	the intervention setting X to the value x
$M = \langle U, V, F, P(U) \rangle$	a structural causal model
r, p, e	reference, perception, control error ($e = r - p$)
k	loop gain (the loop is stable for $0 < k < 2$)
s	structural grounding score, $s \in [0, 1]$
$g(s)$	attenuation map; $g(0) = 0$, monotone, $g(s) \in [0, 1]$
$O1 < O2 < O3$	the strict rung order of the inference operators
PN, PS	probabilities of necessity and sufficiency

Appendix F — Casebook

Twelve non-normative cases, chosen to cover the distinct ways a claim can pass or fail at the analytical resolution. Each gives the output, its configuration, the verdict, and the correction.

F.1 Confounded policy claim. “The new curriculum raised test scores.” Configuration ($L6 \times C5 \times Q3 \cdot O2$). Schools that adopted it differ in funding and intake, an unblocked back-door. Supported: O1. Verdict: flag (confounding). Correction: name an adjustment set, or retreat to “adoption is associated with higher scores.”

F.2 Collider / selection. “Among admitted patients, smokers had milder disease, so smoking protects.” Configuration ($L1 \times C5 \times Q3 \cdot O2$). Conditioning on admission — a common effect of smoking and severity — induces the association. Verdict: flag (collider). Correction: do not condition on admission; estimate in the unselected population.

F.3 Reverse causation. “Umbrella sales predict rain, so selling umbrellas brings rain.” Configuration ($L6 \times C5 \times Q3 \cdot O2$). The association is real; the arrow runs from rain to sales. Verdict: flag (direction). Correction: fix the direction with temporal or interventional evidence.

F.4 Licensed experiment. “In the randomized trial, the drug lowered mortality.” Configuration ($L1 \times C5 \times Q3 \cdot O2$). Randomization severs the arrows into treatment. Supported: O2. Verdict: pass.

F.5 Counterfactual without a model. “But for the delay, the deal would have closed.” Configuration ($L6 \times C5 \times Q3 \cdot O3$). No structural model supports the abduction. Supported: at most O1. Verdict: flag (severe). Correction: build the model or state the modest associational claim.

F.6 Licensed counterfactual. “In the validated simulator, had the valve stayed open, pressure would not have spiked.” Configuration ($L1 \times C5 \times Q3 \cdot O3$). A specified mechanism supports abduction-action-prediction. Supported: O3. Verdict: pass, with the model logged as the load-bearing premise.

F.7 Under-claim. “Projects using the flag report fewer bugs,” when a randomized rollout licenses the causal claim. Configuration ($L4 \times C5$

× Q3 · O1). Asserted O1, supported O2. Verdict: logged under-claim (not attenuated; warrant wasted). Correction: state the earned causal claim.

F.8 Definition read causally. “Being even causes a number to be divisible by two.” Configuration (L4 × C6 × Q4). The relation is definitional, not causal. Verdict: flag (category error). Correction: state it as a definitional equivalence, not an intervention.

F.9 Subjective claim, causal phrasing. “Liking the design makes it good.” Configuration (L5 × C4 × Q1). A preference (subjective) is cast as producing an objective property. Verdict: flag (domain boundary). Correction: separate the subjective preference from any objective criterion of quality.

F.10 Transport beyond support. “The effect found in the trial will hold in this very different population.” Configuration (L6 × C5 × Q3 · O2). The identifying assumptions may not transfer. Verdict: flag (extrapolation). Correction: state and defend a transport assumption, or restrict the claim to the studied population.

F.11 Composite sentence. “The popular reform will obviously cut crime.” Decomposes into a Q1/O1 popularity claim, a Q3/O2 crime prediction, and a modal overreach. Verdict: pass the parts that hold, attenuate the unearned causal-necessity. Correction: assert the components separately.

F.12 Over-adjustment. “Adjusting for post-treatment engagement, the campaign had no effect.” Configuration (L6 × C5 × Q3 · O2). Engagement is a mediator on the causal path; conditioning on it removes part of the effect. Verdict: flag (over-adjustment). Correction: do not adjust for mediators when estimating a total effect.

F.13 Instrumental variable. “Using distance to college as a lever, schooling raises earnings.” Configuration (L6 × C5 × Q3 · O2). An instrument is licensed only if it affects the outcome solely through the cause and is itself unconfounded. Verdict: pass only if those assumptions are stated; otherwise flag.

F.14 Front-door mediation. “The ad raised sales through store visits, though buyer wealth is unobserved.” Configuration (L6 × C5 × Q3 · O2). A front-door claim, licensed if visits fully mediate and are

shielded from the confounder. Verdict: pass with the mediator and shielding declared.

F.15 Base-rate neglect. “The test is 99% accurate and positive, so the disease is almost certain.” Configuration ($L4 \times C5 \times Q3 \cdot O1$). This is a conditioning error, not a causal one: it ignores the prior. Verdict: flag ($O1$ mis-computed); apply Bayes’ rule with the base rate.

F.16 Selection on the outcome. “Successful founders dropped out, so dropping out causes success.” Configuration ($L6 \times C5 \times Q3 \cdot O2$). Sampling on success is collider selection. Verdict: flag; estimate over all founders, not the successful ones.

F.17 Necessity versus sufficiency. “The spark caused the fire.” Configuration ($L1 \times C5 \times Q3 \cdot O3$). A but-for claim; fuel and oxygen were also necessary. Verdict: pass as a contributing necessary cause only if the model supports the probability of necessity; do not read sufficiency into it.

F.18 Mechanism at the substrate. “Raising temperature increases the reaction rate.” Configuration ($L1 \times C5 \times Q3 \cdot O2$). A well-characterized physical mechanism. Supported: $O2$. Verdict: pass.

F.19 Meta over-generalization. “The model with more parameters is always better.” Configuration ($L4 \times C6 \times Q4$). A meta-level claim asserted as necessary; it is false as stated and is not a causal operation. Verdict: flag (meta over-generalization); restate with conditions.

F.20 Correct ascent via mediation. “The drug lowers mortality by lowering blood pressure, shown in trial and mechanism.” Configuration ($L1 \times C5 \times Q3 \cdot O2$). Both the total effect and the mediating mechanism are evidenced. Verdict: pass; the mediation claim is licensed.

Glossary

Analytical configuration — A four-tuple (Layer $\times$ Category $\times$ Quadrant $\cdot$ Operator); the unit on which the 864 engine acts.

Conditioning (01) — The associational operator, $P(Y | X)$; the first rung of the causal hierarchy.

Intervention / do-operator (02) — The interventional operator, $P(Y | \text{do}(X))$; the effect of setting rather than observing X .

Counterfactual (03) — The operator of what-would-have-been, $P(Y_x | X', Y')$; the third rung, requiring a structural causal model.

Structural causal model — Functions relating each variable to its parents plus a distribution over exogenous noise; the object counterfactuals require.

Back-door criterion — A graphical condition on a set Z that licenses identifying $P(Y | \text{do}(X))$ by adjustment.

Causal Hierarchy Theorem — Pearl's result that the three rungs do not collapse: higher-rung facts are not determined by lower-rung data alone.

Reference signal (04) — The externally set target value of a controlled perception; the grounding target.

Comparator / control error (05) — The element forming $e = r - p$ and the signed gap it produces.

Predictive coding — A scheme minimizing prediction error / free energy; an admissible alternative formalization of the comparator.

Output attenuation (06) — The actuator output $= g(s) \cdot \text{raw_output}$, with $g(0)=0$, g monotone, $g(s) \in [0,1]$.

Grounding score (s) — A value in $[0,1]$ measuring how positioned and operator-supported a candidate output is; 1 at zero control error.

Zero-Incidence Attenuation — The property that $s = 0$ forces output $= 0$: a fully ungrounded output is not emitted.

Unsupported Causality — Asserting an operator above the rung the evidence supports; the chief failure of the analytical resolution.

Operator accuracy — The requirement that the correct operator be applied to the correct claim; over-claiming and under-claiming are its two errors.

Pre-decode — Acting on the generative probability or logit vector before token selection, so attenuation shapes generation.

d-separation — A graphical criterion for reading conditional independence off a causal graph; the basis of identification.

Confounder — A common cause of X and Y that biases the observed association as an estimate of effect.

Collider — A common effect of X and Y; conditioning on it induces a spurious association (selection bias).

Front-door criterion — An identification route through an unconfounded mediator when no back-door set is available.

Exogenous / endogenous — Background variables (U) set outside the model versus variables (V) determined within it.

Loop gain (k) — The factor governing how fast the control error decays; stable for $0 < k < 2$.

Probability of necessity / sufficiency — Counterfactual measures (PN, PS) of but-for and bring-about causation; often only bounded from data.

References

- Amari, S. (2016). *Information Geometry and Its Applications*. Springer.
- Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.
- Habermas, J. (1984). *The Theory of Communicative Action*, Vol. 1. Beacon Press.
- Hara, S. (2026). *The HNS Origin Trilogy*. Natural Structure Works.
- Hara, S. (2026). *Human Natural Structure: Foundational & Observational Specification*. Natural Structure Works.
- Kahneman, D. (2011). *Thinking, Fast and Slow*. Farrar, Straus and Giroux.
- Miller, E. K., & Cohen, J. D. (2001). An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience*, 24, 167–202.
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.
- Pearl, J., & Mackenzie, D. (2018). *The Book of Why*. Basic Books.
- Powers, W. T. (1973). *Behavior: The Control of Perception*. Aldine.
- Rao, R. P. N., & Ballard, D. H. (1999). Predictive coding in the visual cortex. *Nature Neuroscience*, 2(1), 79–87.
- Smith, B. (2015). *Basic Formal Ontology (BFO) 2.0 Specification and User's Guide*.
- Spirtes, P., Glymour, C., & Scheines, R. (2000). *Causation, Prediction, and Search* (2nd ed.). MIT Press.
- Wiener, N. (1948). *Cybernetics: or Control and Communication in the Animal and the Machine*. MIT Press.